

Nihar Domala

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EXPERIENCE

Capital One (Contract)

McLean, USA

Data Engineer

Jun 2025 – Present

- Built PySpark/AWS Glue pipeline ingesting 4.2B+ daily records into Delta Lake, powering marketing campaigns, ML models, and analytics for 14 downstream consumers – structured on Medallion Architecture
- Developed internal Lambda API tooling that automated pipeline configuration, testing, and deployment workflows, reducing engineering effort by ~80%
- Engineered 5+ decisioning DMNs integrated across 3 AWS Step Functions workflows, enabling 5x increase in concessionary payment plan presentments
- Automated end-to-end test account conditioning pipeline by chaining 11+ Python and shell script steps across internal tool APIs and DynamoDB, cutting conditioning time ~80% and eliminating manual intervention
- Diagnosed and resolved 15+ high-severity data pipeline incidents via root cause analysis, eliminating recurring SLA breaches across 14 downstream data consumers
- Delivered 10+ enterprise Delta Lake Big Table integrations within strict delivery SLAs by coordinating requirements with downstream teams and authoring SQL transformation configs and schema validation logic in Databricks

Samsung R&D Institute

Bangalore, India

Software Development Engineer Intern

May 2022 – Jul 2022

- Identified top 30% high-data-consumption locations for Samsung store installations by training a K-Means clustering model on 3,000+ location records, benchmarking against 4+ algorithms, with real-time Plotly dashboards for site selection

iQuanti Inc.

Bangalore, India

Machine Learning Engineer Intern

May 2021 – Jul 2021

- Led team of 3 to build NLP preprocessing pipelines (TF-IDF vectorization, GloVe tokenization, BERT tokenization) for 47,684 imbalanced search queries across 5 marketing funnel intent classes
- Benchmarked Naive Bayes (TF-IDF), LSTM (GloVe 300-dim), and BERT (110M params) on 5-class search intent classification, achieving 96.78% validation accuracy using TensorFlow and scikit-learn

SKILLS

Programming & Tools: Python, SQL, Linux/Unix, Git, Jenkins, Docker, Terraform, GitHub Actions

Databases: MySQL, Snowflake, MongoDB, PostgreSQL, Neo4j, BigQuery

Big Data & Visualization: PySpark, Spark, Delta Lake, dbt, Databricks, Airflow, Kafka, Hadoop, QuickSight

ML & AI: PyTorch, scikit-learn, HuggingFace Transformers, MLflow, BERT, LSTM, CNNs

AWS Services: Lambda, Glue, S3, DynamoDB, Redshift, Kinesis, IAM, EMR, Step Functions, Athena, EC2

PROJECTS

CloudMart: MultiCloud AI-powered E-Commerce Cloud Application

Jan 2025 – Feb 2025

- Deployed MultiCloud AI e-commerce platform on AWS (EKS, Bedrock, Terraform, CodePipeline), GCP (BigQuery, Looker Studio), and Azure (Sentiment Analysis); cut customer support costs 90% via Claude Bedrock LLM assistant with RAG-based product catalog retrieval and enabled real-time sales analytics

PUBLICATIONS

Bonferroni Mean-Based Pooling Function in CNNs | *IEEE INDICON 2023 – MV Chauhan Best Paper Award*

- Implemented novel mean-based pooling aggregation in CNN layers using PyTorch; outperformed classical pooling on state-of-the-art architectures across real medical datasets

AWARDS & HONORS

GWU Datathon 2024: Secured 1st place, winning \$400 for predictive modeling of hotel booking patterns

Global Leaders Award: Awarded \$17,000 tuition fellowship by GWU in recognition of academic excellence

EDUCATION

The George Washington University

Washington DC, US

Master of Science in Data Science | GPA: 3.97/4.0

Aug 2023 – May 2025

Indian Institute of Technology, Kharagpur

West Bengal, India

Bachelor of Technology in Instrumentation Engineering | Micro-specialization in AI & Applications

Jul 2019 – May 2023